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Enhanced account security monitoring

Abstract

A security monitoring system is a network of distributed computing systems. The distributed computing systems include an ingestion & rules engine, a compromised identity set service, and a message bus. The ingestion & rules engine, the compromised identity set service, and the message bus may be separated in the security monitoring system geographically from any other distributed computing system in the security monitoring system. The security monitoring system may communicate electronically with an information distributor, an identity system, and external services.

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Background/Summary

BACKGROUND

(1) Electronic data repositories may store user personal data. This data may be used by computer systems to authenticate a user for providing access to data or services associated with the computer systems.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings, which are incorporated in and form a part of this specification, illustrate examples of the disclosure and, together with the description, explain principles of the examples. In the drawings, like reference symbols and numerals indicate the same or similar components.
- (2) FIG. 1A illustrates a security monitoring system.
- (3) FIG. 1B illustrates the security monitoring system in a distributed computing environment.
- (4) FIG. 2 illustrates a signal flow diagram for the security monitoring system.
- (5) FIG. 3A is a flowchart that illustrates an example of processing for an ingestion & rules engine.
- (6) FIG. 3B is a flowchart that illustrates an example of processing for a compromised identity set service.
- (7) FIG. 3C is a flowchart that illustrates an example of processing for a message bus.
- (8) Like elements in the various figures are denoted by like reference numerals for consistency.

DETAILED DESCRIPTION

(9) Hereinafter, exemplary embodiments are described in detail with reference to the accompanying drawings. For the sake of clarity and conciseness, matters related to the present embodiments that are well known in the art have not been described.

(10) Embodiments of the disclosure are described in detail below with reference to the accompanying figures. Unless otherwise indicated, like parts and method steps are referred to with like reference numerals.

(11) Electronic data repositories may store user personal data, including a user's name, address, device information, credit card information, birthdate, and the like. This data may be used by computer systems to authenticate a user for providing access to data or services associated with the computer systems. As data breaches exposing personal data of data repositories become more common, particularly given the ease with which malicious actors can obtain the data, the widespread abuse of compromised personal data (including user credentials) will only increase. Users and companies can do little to prevent malicious use of data once exposed, leaving consumers open to fraud and companies to experience unauthorized access to systems and financial loss due to that fraud. While there are known repositories of users' account information and passwords exposed on the public Internet and the dark web, vast amounts of other personal data can also be utilized for nefarious purposes. For example, an ecommerce site could use the last four digits of a credit card for verification for purchase authorization, or a healthcare site could use the last four digits of a social security number as a form of verification to make account changes. When this type of information is made available as the result of a data breach, the ecommerce or healthcare site may not know the specifics of a breach and accept this information as a valid form of verification and allow a process or transaction to continue.

(12) Embodiments provided herein address these and other technical issues associated with data breaches of personal information in electronic data repositories. For example, embodiments provided herein enable, among other things, identification of personal information involved in a data breach and electronic notification of impacted users and systems.

(13) FIG. 1A illustrates a security monitoring system **11**. The security monitoring system **11** is a network of distributed computing systems. The distributed computing systems include an ingestion & rules engine **112**, a compromised identity set service **114**, and a message bus **116**. The ingestion & rules engine **112**, the compromised identity set service **114**, and the message bus **116** in the security monitoring system **11** may be separated geographically from any other distributed computing system in the security monitoring system **11**. The ingestion & rules engine **112**, the compromised identity set service **114**, and the message bus **116** may perform their tasks autonomously. The security monitoring system **11** may communicate electronically with an information distributor **12**, an identity system **13**, and external services **14**.

(14) FIG. 1B illustrates the security monitoring system **11** in a distributed computing environment **16**. A communication network **18** may include a public or private data network. The data network may comprise or be part of a data bus, a wired or wireless information network, a public switched telephone network, a satellite network, a local area network (LAN), a wide area network (WAN), and/or the Internet. In the distributed computing environment **16**, the ingestion & rules engine **112**, the compromised identity set service **114**, and the message bus **116** may simultaneously perform divided tasks in parallel with each other.

(15) Due at least in part to the ingestion & rules engine **112** being operable in the distributed computing environment **16**, a human is unable to perform the functions of the ingestion & rules engine **112**.

(16) Due at least in part to the compromised identity set service **114** being operable in the distributed computing environment **16**, a human is unable to perform the functions of the compromised identity set service **114**.

(17) Due at least in part to the message bus **116** being operable in the distributed computing environment **16**, a human is unable to perform the functions of the message bus **116**.

(18) The ingestion & rules engine **112** may be a computing apparatus that includes software and hardware. The hardware for the ingestion & rules engine **112** may be a platform that resides at a physical, geographic location such as a country, state, county, city, or building. The ingestion & rules engine **112** may be a cloud computing platform that resides in a cloud based architecture without a designated geographic location.

(19) The hardware for the ingestion & rules engine **112** may include an ingestion & rules engine (IRE) processor **122** and a database **132**. The database **132** may be non-transitory processor readable memory containing the software for the ingestion & rules engine **112**.

(20) The software for the ingestion & rules engine **112** may include, for example, firmware, one or more applications, program data, filters, rules, one or more program modules, and other executable instructions. The software for the ingestion & rules engine **112** may be an application programming interface that permits the ingestion & rules engine **112** to electronically communicate with the database **132**, the identity system **13**, and the compromised identity set service **114**. The application programming interface for the ingestion & rules engine **112** may be a program that when, executed, causes the IRE processor **122** to perform the processing for the ingestion & rules engine **112**. The IRE processor **122** is an electronic processor.

(21) The ingestion & rules engine **112** includes the database **132**. The database **132** may include read-only memory ("ROM"), random access memory ("RAM"), other non-transitory computer-readable media, or a combination thereof. The database **132** may include the instructions that are executable by the IRE processor **122**. Information stored in the database **132** is accessible to the ingestion & rules engine **112**, the compromised identity set service **114**, and the message bus **116**.

(22) FIG. 2 illustrates a signal flow diagram for the security monitoring system **11**. With reference to FIG. 2, the ingestion & rules engine **112** is a processing engine that extracts and processes data entries from data breach sets **206** to identify user account information existing in the identity system **13** that is to be considered compromised or leaked. The data breach set **206** may be electronic information from the information distributor **12**. The information distributor **12** may be a third party information provider. The information distributor **12** may be multiple third party information providers. Each third party information provider may be a government entity, a software-as-a-service (SaaS) provider, a security vendor, a secure aggregator, or any other entity that collects and distributes the data breach sets **206**.

(23) The ingestion & rules engine **112** may receive the data breach sets **206** from the information distributor **12**. The data breach sets **206** may include records containing breached data. The breached data is information that is personal, sensitive, private, and/or proprietary. The breached data is obtained, appropriated and/or altered by an entity other than an owner of the information without prior authorization from the owner of the information. This breached data may include compromised information from a particular data breach. The compromised information may include usernames, passwords, addresses, credit card information, banking information, and/or any other information that is personal, sensitive, private, and/or proprietary.

(24) The ingestion & rules engine **112** may communicate, directly or indirectly, with the information distributor **12**. A firewall in the ingestion & rules engine **112** may prevent the information distributor **12** from accessing the database **132**, the identity system **13**, the compromised identity set service **114**, and the message bus **116**.

(25) The ingestion & rules engine **112** may receive, from the information distributor **12**, a data notification **202** that informs the ingestion & rules engine **112** of a new data breach set **206** for output by the information distributor **12**. To receive the data breach sets **206** from the information distributor **12**, the ingestion & rules engine **112** may issue a data request **204** to the information distributor **12** for the data breach sets **206**. The ingestion & rules engine **112** may issue the data request **204** to the information distributor **12** as a response to the ingestion & rules engine **112** receiving the data notification **202**. In some examples, the ingestion & rules engine **112** may receive the data breach sets **206** in the absence of an issuance of the data request **204** from the

ingestion & rules engine **112**. When the ingestion & rules engine **112** receives the data breach sets **206**, the ingestion & rules engine **112** may issue a record request **208** to the identity system **13** for user account information.

(26) The identity system **13** may be a computing apparatus that includes software and hardware. The hardware for the identity system **13** may be a platform that resides at a physical, geographic location such as a country, state, county, city, or building. The identity system **13** may be a cloud computing platform that resides in a cloud-based architecture without a designated geographic location. The hardware for the identity system **13** may include non-transitory processor readable memory containing the software for the identity system **13**.

(27) The software for the identity system **13** may include, for example, firmware, one or more applications, program data, filters, rules, one or more program modules, and other executable instructions. The instructions may include software executable by an electronic processor to enable the identity system **13** to, among other things, perform or control the functions of the identity system **13** described herein.

(28) The software for the identity system **13** may be an application programming interface that permits the identity system **13** to electronically communicate with the ingestion & rules engine **112**. The application programming interface for the identity system **13** may be a program that when, executed, causes an electronic processor to perform the processing for the identity system **13**. The instructions may include software executable by the electronic processor to enable the identity system **13** to, among other things, perform or control the functions of the identity system **13** described herein during a predetermined time period.

(29) The ingestion & rules engine **112** may receive the data breach sets **206** intermittently from the information distributor **12**. The ingestion & rules engine **112** may receive the data breach sets **206** from the information distributor **12** on a schedule or a timetable.

(30) The ingestion & rules engine **112** may extract, from the data breach sets **206**, breached data pertaining to system users. A user may be a consumer, individual or entity that may purchase or use a product and/or service. The user is a system user during the time period that the user account information for the user is present in the identity system **13**. The ingestion & rules engine **112** may convert the breached data into data entries when the ingestion & rules engine **112** extracts the breached data from the data breach sets **206**. When converting the breached data into the into data entries, the ingestion & rules engine **112** may filter out information in the breached data that is not particularly relevant or harmful to the security of the system users. The ingestion & rules engine **112** may store the data entries into the database **132** after filtering out the information. By filtering out the information when the ingestion & rules engine **112** converts the breached data into the data entries, the ingestion & rules engine **112** may reduce the amount of storage space in the database **132** that is required to store the data entries in the database **132**. The ingestion & rules engine **112** may store, into the database **132**, the data entries in a format that is compatible with the records in the identity system **13**.

(31) The identity system **13** is a processing engine that receives, stores, maintains, manages, and safeguards the user account information associated with the system users. The user account information may include authentication information and profile information for each of the system users. The authentication information may include usernames, passwords, certificates, and/or any other information that is personal, sensitive, private, and/or proprietary to the system users. The user profile information may include names, addresses, phone numbers, and/or any other information that identifies the system users.

(32) The ingestion & rules engine **112** may identify the user account information that is compromised. The user account information may become compromised user information when the ingestion & rules engine **112** detects at least a portion of the user account information in the data entries. The ingestion & rules engine **112** may designate the user account information as uncompromised user information when the ingestion & rules engine **112** fails to detect at least a

portion of the user account information in the data entries. The ingestion & rules engine **112** may detect any user account information existing in the data entries when the ingestion & rules engine **112** compares the user account information to the data entries.

(33) The identity system **13** may store the user account information for a system user in the user data record **210**. The ingestion & rules engine **112** may compare the user data record **210** to the data entries to detect the breached data in the user account information for the system user. By comparing the user data record **210** to the data entries, the ingestion & rules engine **112** may determine whether or not the user data record **210** includes compromised user information.

(34) The ingestion & rules engine **112** may schedule comparisons to occur at regular time intervals with a same amount of time between the comparisons. Alternatively, the ingestion & rules engine **112** may compare the user data record **210** to the data entries immediately upon receipt, by the ingestion & rules engine **112**, of the data breach set **206** from the information distributor **12**.

(35) The ingestion & rules engine **112** may send, to the identity system **13**, the record request **208** for the user data record **210** when the ingestion & rules engine **112** receives the new the data breach set **206** from the information distributor **12**. When the identity system **13** receives the record request **208**, the identity system **13** may output the user data record **210** to the ingestion & rules engine **112**. Alternatively, the identity system **13** may, without receiving the record request **208** from the ingestion & rules engine **112**, output the user data record **210** to the ingestion & rules engine **112** when the identity system **13** updates the user data record **210**.

(36) Upon receipt of the user data record **210** from the identity system **13**, the ingestion & rules engine **112** may extract the data entries from the database **132**. In some instances, the ingestion & rules engine **112** may extract the data entries from the database **132** prior to the ingestion & rules engine **112** receiving the user data record **210** from the identity system **13**.

(37) When the ingestion & rules engine **112** receives the user data record **210** from the identity system **13** and extracts the data entries from the database **132**, the ingestion & rules engine **112** may compare the user account information for the system user with the breached data in the data entries to ascertain whether or not some of the user account information in the user data record **210** is present in at least one the data entries. By comparing the user data record **210** to the breached data, the ingestion & rules engine **112** may identify the breached data pertaining to a system user in the data entries.

(38) The ingestion & rules engine **112** may designate the user data record **210** as being compromised when the ingestion & rules engine **112** determines that at least some of the user account information in the user data record **210** is present in at least one of the data entries.

(39) When the ingestion & rules engine **112** determines that at least some of the user account information in the user data record **210** is present in the data entries, the ingestion & rules engine **112** may control the identity system **13** in a manner that causes the identity system **13** send an advisory to the system user. The advisory is a recommendation for the system user to update the user account information in the user data record **210** in a manner that may remove the breached data from the user data record **210**.

(40) When the ingestion & rules engine **112** determines that at least some of the user account information in the user data record **210** is present in the data entries, the ingestion & rules engine **112** may perform a risk assessment. The ingestion & rules engine **112** may execute a risk algorithm to perform the risk assessment. The risk algorithm may be a machine learning algorithm. The machine learning algorithm may be a neural network model. The risk algorithm may be a heuristic algorithm. The ingestion & rules engine **112** may identify, in the data entries, the breached data pertaining to a system user. The user account information in the user data record **210** and the breached data pertaining to the system user are inputs to the risk algorithm.

(41) When the ingestion & rules engine **112** executes the risk algorithm, the ingestion & rules engine **112** may calculate a value for a risk score. The risk score quantifies, for the security monitoring system **11**, an amount of risk to the system user whose user account information is

compromised and/or an amount of risk to entities or systems that interact with the system user. The types of risk to the system user may include theft of the system user's identity, damage to the reputation of the system user, financial damage to the system user, and/or other risks that are associated with the user data record **210** being compromised.

(42) The risk score may quantify the severity of the of risk to the system user whose user account information is compromised. To quantify the amount of risk, the risk score may be represented by a scale that ranges from 0 to 100. For example, a risk score of 0 may suggest an absence of risk to the system user as a result of the user data record **210** being compromised. A risk score of 100 may suggest the highest amount of risk to the system user as the result of the user data record **210** being compromised.

(43) The ingestion & rules engine **112** may generate notification information. The notification information may include identity information that uniquely identifies the system user. The notification information may also include the risk score. The notification information may include a reason for the notification information.

(44) The ingestion & rules engine **112** may generate the notification information when the identity system **13** detects the user account information in the user data record **210** in at least one the data entries. The user account information is compromised when appropriated without authorization by the system user. The user data record **210** is compromised when the user account information in the user data record **210** is present in at least one the data entries.

(45) The reason for the notification information may be the detection, made by the ingestion & rules engine **112**, of the user account information in the user data record **210** in at least one the data entries. When the detection is the reason, the notification information may include information that highlights the user account information in the user data record **210** that is determined by the ingestion & rules engine **112** to be compromised.

(46) The ingestion & rules engine **112** may generate the notification information when the identity system **13** creates the user data record **210**. The identity system **13** may include a user interface that permits the identity system **13** to communicate with a user. The user interface may permit the user to create the user data record **210**. The user is a new system user upon the instance that the identity system **13** creates the user data record **210** for the user in the absence of any other the user data record **210** for the user in the identity system **13**. The reason for the notification information may be the storage, into the identity system **13**, of user account information for the new system user. When the storage of the user account information for the new system user is the reason, the notification information may include information that identifies the new system user.

(47) When the identity system **13** updates the user data record **210**, the ingestion & rules engine **112** may generate the notification information. The identity system **13** may receive an instruction from the user interface that causes the identity system **13** to update the user account information in the user data record **210**. Updating the user data record **210** modifies the user account information in the user data record **210**. The reason for the notification information may be the modification of the user data record **210**. When the updating the user data record **210** is the reason, the notification information may include the identity information along with information that pertains to the modification.

(48) When the identity system **13** deletes the user data record **210** from the identity system **13**, the ingestion & rules engine **112** may generate the notification information. The identity system **13** may receive an instruction from the user interface that causes the identity system **13** to delete, from the identity system **13**, the user data record **210** for the system user. Deletion of the user data record **210** for the system user removes, from the identity system **13**, the user account information for the system user. The reason for the notification information may be the deletion of the user data record **210** for the system user from the identity system **13**. When the deletion is the reason, the notification information may include the identity information along with information that pertains to the deletion.

(49) The ingestion & rules engine **112** may package the notification information in an information record **214** when the ingestion & rules engine **112** generates the notification information. When the ingestion & rules engine **112** packages the notification information in the information record **214**, the ingestion & rules engine **112** may output the information record **214** to the compromised identity set service **114**. For enhanced security of the information record **214**, the ingestion & rules engine **112** may encrypt the information record **214** before outputting the information record **214** as an encrypted the information record **214**. The information record **214** may comprise information that is exclusively for the system user.

(50) When the ingestion & rules engine **112** outputs the notification information in the information record **214**, the ingestion & rules engine **112** may output the information record **214** immediately when the ingestion & rules engine **112** generates the notification information. Alternatively, when the ingestion & rules engine **112** generates the notification information, the ingestion & rules engine **112** may output the information record **214** at a time slot within a time period. For example, the ingestion & rules engine **112** may output the information record **214** only at a specific time during a day.

(51) The ingestion & rules engine **112** may collect notification information for the system users in a batch of the information records **214**. The batch may include a subset of the user data records **210** that are in the identity system **13**. The subset may include the user data records **210** in which the user account information those user data records **210** is present in in at least one the data entries. Each record in the batch may be for a system user. For example, one information record **214** in the batch may include a notification information for one of the system users whereas another information record **214** in the batch may include a notification information for another of the system users. When the ingestion & rules engine **112** collects the notification information in the batch, the ingestion & rules engine **112** may output the batch to the compromised identity set service **114**. The ingestion & rules engine **112** may output the batch immediately when the ingestion & rules engine **112** collects the notification information in the batch. Alternatively, when the ingestion & rules engine **112** collects the notification information in the batch, the ingestion & rules engine **112** may output the batch at a time slot within a time period. For example, the ingestion & rules engine **112** may output the batch only at a specific time during a day.

(52) Due at least in part to the massive amount of information in the data breach sets **206** for the ingestion & rules engine **112** to assess and process, a human is unable to perform the functions of the ingestion & rules engine **112**.

(53) The compromised identity set service **114** may be a computing apparatus that includes software and hardware. The hardware for the compromised identity set service **114** may be a platform that resides at a physical, geographic location such as a country, state, county, city, or building. The compromised identity set service **114** may be a cloud computing platform that resides in a cloud based architecture without a designated geographic location.

(54) The hardware for the compromised identity set service **114** may include a compromised identity set service (CISS) processor **124** and a dataset **134**.

(55) The dataset **134** may be non-transitory processor readable memory containing the software for the compromised identity set service **114**. The software for the compromised identity set service **114** may include, for example, firmware, one or more applications, program data, filters, rules, one or more program modules, and other executable instructions. The software for the compromised identity set service **114** may be an application programming interface that permits the compromised identity set service **114** to electronically communicate with the dataset **134**, the ingestion & rules engine **112**, the message bus **116**, and the external services **14**. The external services **14** may include the data services **142**, the partner app shopping **144**, and the security subscription services **146**.

(56) The application programming interface for the compromised identity set service **114** may be a program that when, executed, causes the CISS processor **124** to perform the processing for the

compromised identity set service **114**. The CISS processor **124** is an electronic processor.

(57) The compromised identity set service **114** may include the dataset **134**. The dataset **134** may include read-only memory (“ROM”), random access memory (“RAM”), other non-transitory computer-readable media, or a combination thereof. The dataset **134** may include the instructions that are executable by the CISS processor **124**. Information stored in the dataset **134** is accessible to the compromised identity set service **114**, the ingestion & rules engine **112**, and the message bus **116**.

(58) The compromised identity set service **114** may communicate, directly or indirectly, with the ingestion & rules engine **112** and the external services **14**. A firewall in the compromised identity set service **114** may prevent the information distributor **12** and the external services **14** from accessing the dataset **134** and the ingestion & rules engine **112**.

(59) The compromised identity set service **114** may communicate, directly or indirectly, with the ingestion & rules engine **112**. The ingestion & rules engine **112** may output the information record **214** to the compromised identity set service **114** in response to an information request **212** from the compromised identity set service **114** for the information record **214**. Alternatively, the ingestion & rules engine **112** may output the information record **214** to the compromised identity set service **114** in the absence of the information request from the compromised identity set service **114** for the information record **214**.

(60) The compromised identity set service **114** may receive, from the ingestion & rules engine **112**, the information record **214** for the system user. The compromised identity set service **114** may receive the information record **214** as a single information record **214** or in the batch of information records **214**. The compromised identity set service **114** may obtain the information record **214** from the batch of information records **214** when the compromised identity set service **114** receives the batch from the ingestion & rules engine **112**. When the compromised identity set service **114** receives the information record **214** in the form of the encrypted the information record **214**, the compromised identity set service **114** may decrypt the information record **214** before obtaining the notification information from the information record **214**.

(61) The dataset **134** is a data repository that stores identity files for the system users. An identity file in the dataset **134** may comprise information that is exclusively for the system user. The compromised identity set service **114** may extract, from the information record **214**, the notification information for the system user when the compromised identity set service **114** receives the information record **214**. When the notification information is for the new system user, the compromised identity set service **114** may create, in the dataset **134**, an identity file for the new system user. The compromised identity set service **114** may obtain, from the notification information, the identity information that uniquely identifies the system user. When the compromised identity set service **114** stores the notification information in the identity file, the compromised identity set service **114** may index the identity file in the dataset **134** by the identity information.

(62) The compromised identity set service **114** may process the notification information to determine whether or not the reason for the notification information is the detection of the user account information for the system user in at least one the data entries from the data breach sets **206**. When the compromised identity set service **114** determines that the reason for the notification information is the detection of the user account information for the system user in at least one the data entries from the data breach sets **206**, the compromised identity set service **114** may output, to the message bus **116**, the communication **216** that notifies the message bus **116** of the compromised information in the user account information for the system user. The communication **216** may include the identity information that uniquely identifies the system user. The communication **216** may also include the risk score that quantifies the amount of risk to the system user. The communication **216** may include the reason for the notification information.

(63) The message bus **116** may broadcast the communication **216** to the security subscription

services **146** in a publication **218**. The publication **218** may include the notification informing the security subscription services **146** that the user account information for the system user is compromised. The message bus **116** may broadcast the publication **218** to the security subscription services **146** in the absence of a request, from any of the security subscription services **146**, for the publication **218**. The publication **218** from the message bus **116** is unprompted.

(64) For enhanced security of the publication **218**, the message bus **116** may encrypt the communication **216** before broadcasting the communication **216** in an encrypted publication **218**. When broadcasting the publication **218** to the security subscription services **146**, the message bus **116** may broadcast the communication **216** to all of the security subscription services **146** simultaneously in a single publication **218**.

(65) The message bus **116** may broadcast the publication **218** immediately when the message bus **116** receives the communication **216** from the compromised identity set service **114**. Alternatively, the message bus **116** may broadcast the publication **218** at a time slot within a time period. For example, the message bus **116** may broadcast the publication **218** only at a specific time during a day.

(66) The message bus **116** may collect a plurality of the communications **216** in a batch of publications **218**. Each communication **216** in the batch of publications **218** may be for a separate and distinct system user. For example, a communication **216** in the batch of publications **218** may notify the message bus **116** of the compromised information in the user account information for one of the system users whereas another communication **216** in the batch of publications **218** may notify the message bus **116** of the compromised information in the user account information for another of the system users. When the message bus **116** collects the communications **216** in the batch of publications **218**, the message bus **116** may broadcast the batch of publications **218** to the security subscription services **146**. The message bus **116** may broadcast the batch of publications **218** immediately when the message bus **116** collects the communications **216** in the batch of publications **218**. Alternatively, when the message bus **116** collects the communications **216** in the batch of publications **218**, the message bus **116** may broadcast the batch of publications **218** at a time slot within a time period. For example, the message bus **116** may broadcast the batch of publications **218** only at a specific time during a day.

(67) The message bus **116** may be a computing apparatus that includes software and hardware. The hardware for the message bus **116** may be a platform that resides at a physical, geographic location such as a country, state, county, city, or building. The message bus **116** may be a cloud computing platform that resides in a cloud based architecture without a designated geographic location.

(68) The hardware for the message bus **116** may include a message bus (MB) processor **126** and memory **136**. The memory **136** may be non-transitory processor readable memory containing the software for the message bus **116**. The software for the message bus **116** may include, for example, firmware, one or more applications, program data, filters, rules, one or more program modules, and other executable instructions. The software for the message bus **116** may be an application programming interface that permits the message bus **116** to electronically communicate with the compromised identity set service **114** and the security subscription services **146**. The application programming interface for the message bus **116** may be a program that when, executed, causes the MB processor **126** to perform the processing for the message bus **116**. The MB processor **126** is an electronic processor.

(69) The message bus **116** includes memory **136**. The memory **136** may include read-only memory (“ROM”), random access memory (“RAM”), other non-transitory computer-readable media, or a combination thereof. The memory **136** may include instructions that are executable by the MB processor **126**. Information stored in the memory **136** is accessible to the ingestion & rules engine **112**, the compromised identity set service **114**, and the message bus **116**.

(70) Due at least in part to the massive amount of information received by the message bus **116** and broadcast from the message bus **116**, a human is unable to perform the functions of the message

bus **116**.

(71) The message bus **116** may communicate, directly or indirectly, with the security subscription service **146**. A firewall in the message bus **116** may prevent the security subscription service **146** from accessing the memory **136** and the ingestion & rules engine **112**.

(72) The compromised identity set service **114** may communicate, directly or indirectly, with the external services **14**. The external services **14** may include the security subscription services **146**, device services, and the partner app shopping **144**.

(73) The compromised identity set service **114** may receive, from an external service **14**, a query **220** to ascertain whether or not the user account information for the system user is compromised. The query **220** may include the identity information that uniquely identifies the system user. When the compromised identity set service **114** receives the query **220**, the compromised identity set service **114** may search the dataset **134** to detect whether or not the identity file for the system user is stored in the dataset **134**.

(74) The compromised identity set service **114** may output a reply **222** to the external service **14** when the compromised identity set service **114** detects the nonexistence of the identity file for the system user in the dataset **134**. The reply **222** notifies the external service **14** of the absence of the breached data in the identity file for the system user. For enhanced security of the reply **222**, the compromised identity set service **114** may encrypt the reply **222** before outputting the reply **222** as an encrypted the reply **222**.

(75) When the compromised identity set service **114** detects the existence of the identity file for the system user in the dataset **134**, the compromised identity set service **114** may process the notification information to determine whether or not the reason for the notification information is the detection of the user account information for the system user in at least one the data entries from the data breach sets **206**. When the compromised identity set service **114** determines that the reason for the notification information is the detection of the user account information for the system user in at least one the data entries from the data breach sets **206**, the compromised identity set service **114** may output, to the external service **14**, the confirmation **222** that notifies the external service **14** of the compromised information in the user account information for the system user. The confirmation **222** may include the identity information that uniquely identifies the system user. The confirmation **222** may include the reason for the notification information. The confirmation **222** may also include the risk score.

(76) Due at least in part to the massive amount of information received by the compromised identity set service **114** and output to the message bus **116**, a human is unable to perform the functions of the compromised identity set service **114**.

(77) The flowchart in FIG. 3A is an example of processing for the ingestion & rules engine **112**. The database **132** may store a program that is executable by the IRE processor **122**. The program, when executed, causes the IRE processor **122** to perform the processing for the ingestion & rules engine **112**. The IRE processor **122** is an electronic processor that may perform processing for the ingestion & rules engine **112** in the distributed computing environment **16**. The IRE processor **122** may perform the processing for the ingestion & rules engine **112** in parallel with the CISS processor **124** performing the processing for the compromised identity set service **114**. The IRE processor **122** may perform the processing for the ingestion & rules engine **112** simultaneously with the CISS processor **124** performing the processing for the compromised identity set service **114**. The IRE processor **122** may perform the processing for the ingestion & rules engine **112** in parallel with the MB processor **126** performing the processing for the message bus **116**. The IRE processor **122** may perform the processing for the ingestion & rules engine **112** simultaneously with the MB processor **126** performing the processing for the message bus **116**.

(78) Processing for the ingestion & rules engine **112** may commence in block **321** of FIG. 3A and advance from block **321** to block **322**. The data notification **202** informs the IRE processor **122** that the new the data breach set **206** from the information distributor **12** is ready for receipt by the

ingestion & rules engine **112**. When the IRE processor **122** detects the absence of the data notification **202** from the information distribution in block **322**, processing in FIG. **3A** advances from block **322** to block **327**. Processing in FIG. **3A** advances from block **322** to block **323** when the IRE processor **122** detects receipt of the data notification **202** in block **322** by the ingestion & rules engine **112** from the information distribution.

(79) In block **323**, the IRE processor **122** may cause the ingestion & rules engine **112** to retrieve the data breach set **206** from the information distributor **12** when the ingestion & rules engine **112** receives the data notification **202** in block **322**. When the ingestion & rules engine **112** retrieves the data breach set **206** from the information distributor **12**, the IRE processor **122** may cause the ingestion & rules engine **112** to store the data breach set **206** in the database **132** in the form of the data entries. Processing in FIG. **3A** advances from block **323** to block **324**.

(80) In block **324**, the IRE processor **122** may access the identity system **13** to retrieve the user data record **210** for the system user. The user data record **210** may comprise the user account information for the system user. The IRE processor **122** may access the database **132** in block **324** to retrieve the data entries. Processing in FIG. **3A** advances from block **324** to block **325**.

(81) In block **325**, the IRE processor **122** may compare the user data record **210** to the data entries to detect the breached data in the user account information for the system user. The user account information may become compromised user information when the ingestion & rules engine **112** detects at least a portion of the user account information in the data entries. The ingestion & rules engine **112** may designate the user account information as uncompromised user information when the ingestion & rules engine **112** fails to detect at least a portion of the user account information in the data entries. By comparing the user data record **210** to the data entries, the IRE processor **122** may determine whether the user data record **210** includes uncompromised user information or includes compromised user information. When the IRE processor **122** determines that the user data record **210** is uncompromised user information, processing in FIG. **3A** advances from block **325** to block **322** and the processing repeats beginning at block **322**. Alternatively, processing in FIG. **3A** advances from block **325** to block **326** when the IRE processor **122** determines that the user data record **210** includes compromised user information.

(82) Also in block **325**, the IRE processor **122** may detect the creation of the user data record **210** when the identity system **13** creates the user data record **210** for the new system user. Processing in FIG. **3A** advances from block **325** to block **326** when the IRE processor **122** detects the creation of the user data record **210**. The IRE processor **122** may detect an update made by the identity system **13** to the user data record **210**. Processing in FIG. **3A** advances from block **325** to block **326** when the IRE processor **122** detects that the identity system **13** has made the update the user data record **210**.

(83) In block **326**, the IRE processor **122** may cause the ingestion & rules engine **112** to output the information record **214** to the compromised identity set service **114**. When the IRE processor **122** causes the ingestion & rules engine **112** to output the information record **214**, processing in FIG. **3A** advances from block **326** to block **322** and the processing repeats beginning at block **322**.

(84) In block **327**, the IRE processor **122** may detect the creation, by the identity system **13**, of the user data record **210**. The IRE processor **122** may detect an update made by the identity system **13** to the user data record **210**. Processing in FIG. **3A** advances from block **327** to block **324** when the IRE processor **122** detects the creation of the user data record **210** or when the IRE processor **122** detects the update to the user data record **210**. In the absence of the IRE processor **122** detecting both a creation of the user data record **210** and an update to the user data record **210**, processing in FIG. **3A** advances from block **327** to block **322** and the processing repeats beginning at block **322**.

(85) The flowchart in FIG. **3B** is an example of processing for the compromised identity set service **114**. The dataset **134** may store a program that is executable by the CISS processor **124**. The program, when executed, causes the CISS processor **124** to perform the processing for the compromised identity set service **114**. The CISS processor **124** is an electronic processor that may

perform processing for the compromised identity set service **114** in the distributed computing environment **16**. The CISS processor **124** may perform the processing for the compromised identity set service **114** in parallel with the IRE processor **122** performing the processing for the ingestion & rules engine **112**. The CISS processor **124** may perform the processing for the compromised identity set service **114** simultaneously with the CISS processor **124** performing the processing for the ingestion & rules engine **112**. The CISS processor **124** may perform the processing for the compromised identity set service **114** in parallel with the MB processor **126** performing the processing for the message bus **116**. The CISS processor **124** may perform the processing for the compromised identity set service **114** simultaneously with the MB processor **126** performing the processing for the message bus **116**.

(86) Processing for the compromised identity set service **114** may commence in block **341** of FIG. **3B** and advance from block **341** to block **342**. The compromised identity set service **114** may receive, from the ingestion & rules engine **112**, the information record **214** for the system user. When the compromised identity set service **114** receives the information record **214**, the CISS processor **124** may designate the information record **214** as a new information record **214**. The CISS processor **124** may designate the information record **214** as the new information record **214** when an identity file for the system user is nonexistent in the dataset **134** or when the CISS processor **124** identifies content the information record **214** that is absent from the identity file for the system user. Processing in FIG. **3B** advances from block **342** to block **343** in the absence of the CISS processor **124** designating the information record **214** as being a new information record **214**.

(87) The compromised identity set service **114** may receive, from an external service **14**, the query **220** to ascertain whether or not the user account information for the system user is compromised. The query **220** may include the identity information that uniquely identifies the system user. When the CISS processor **124** detects the absence in block **343** of receipt of the query **220**, processing in FIG. **3B** advances from block **343** to block **342** and the processing repeats beginning at block **342**. Alternatively, processing in FIG. **3B** advances from block **343** to block **344** when the CISS processor **124** detects receipt of the query **220** in block **343**.

(88) The CISS processor **124** may process the notification information that exists in the identity file for the system user to determine whether or not the reason for generation of the notification information is the detection of the user account information for the system user in at least one the data entries from the data breach sets **206**.

(89) Processing in FIG. **3B** advances from block **344** to block **345** when the CISS processor **124** determines that the reason for the notification information is other than the detection of the user account information for the system user in at least one the data entries from the data breach sets **206**. In block **345**, the CISS processor **124** may cause the compromised identity set service **114** output, to the external service **14** that sent the query **220**, the reply **222** that notifies the external service **14** of the absence of compromised user information in the identity file for the system user. Processing in FIG. **3B** advances from block **345** to block **342** and the processing repeats beginning at block **342**.

(90) Processing in FIG. **3B** advances from block **344** to block **346** when the notification information in the identity file for the system user informs the CISS processor **124** that compromised user information exists in the identity file. In block **346**, the CISS processor **124** may cause the compromised identity set service **114** output, to the external service **14** that sent the query **220**, the confirmation **222** that notifies the external service **14** of the existence of the compromised user information in the identity file for the system user. Processing in FIG. **3B** advances from block **346** to block **342** and the processing repeats beginning at block **342**.

(91) When the CISS processor **124** in block **342** designates the information record **214** received by the compromised identity set service **114** as being a new information record **214**, processing in FIG. **3B** advances from block **342** to block **347**. Processing in FIG. **3B** advances from block **347** to block **342** when the CISS processor **124** identifies the absence of compromised user information in

the identity file for the system user. The processing repeats beginning at block **342**.

(92) Processing in FIG. **3B** advances from block **347** to block **348** when the notification information in the identity file for the system user informs the CISS processor **124** that compromised user information exists in the identity file. In block **348**, the CISS processor **124** may cause the compromised identity set service **114** output, to the message bus **116**, the communication **216** that notifies the message bus **116** of the existence of compromised user information in the identity file for the system user. Processing in FIG. **3B** advances from block **348** to block **343**.

(93) The flowchart in FIG. **3C** is an example of processing for the message bus **116**. The memory **136** may store a program that is executable by the MB processor **126**. The program, when executed, causes the MB processor **126** to perform the processing for the message bus **116**. The MB processor **126** is an electronic processor that may perform processing for the message bus **116** in the distributed computing environment **16**. The MB processor **126** may perform the processing for the message bus **116** in parallel with the IRE processor **122** performing the processing for the ingestion & rules engine **112**. The MB processor **126** may perform the processing for the message bus **116** simultaneously with the IRE processor **122** performing the processing for the ingestion & rules engine **112**. The MB processor **126** may perform the processing for the message bus **116** in parallel with the MB processor **126** performing the processing for the compromised identity set service **114**. The MB processor **126** may perform the processing for the message bus **116** simultaneously with the MB processor **126** performing the processing for the compromised identity set service **114**.

(94) Processing for the message bus **116** may commence in block **361** of FIG. **3C** and advance from block **361** to block **362**. The message bus **116** may receive the communication **216** from the compromised identity set service **114**. Processing in FIG. **3C** remains at block **362** in the absence of the message bus **116** receiving the communication **216** from the compromised identity set service **114**. Processing in FIG. **3C** advances from block **362** to block **363** when the MB processor **126** detects receipt of the communication **216** in block **362**.

(95) In block **363**, the MB processor **126** may cause the message bus **116** to broadcast the publication **218** to the security subscription services **146**. The publication **218** notifies the security subscription services **146** of the existence of compromised user information in the identity file for the system user. When broadcasting the publication **218** to the security subscription services **146**, the message bus **116** may broadcast the publication **218** to all of the security subscription services **146** simultaneously in a single publication **218**. Processing in FIG. **3C** advances from block **363** to block **362** and repeats beginning at block **362**.

(96) Certain operations of methods according to the technology, or of systems executing those methods, can be represented schematically in the figures, or otherwise discussed herein. Unless otherwise specified or limited, representation in the figures of particular operations in particular spatial order can not necessarily require those operations to be executed in a particular sequence corresponding to the particular spatial order. Correspondingly, certain operations represented in the figures, or otherwise disclosed herein, can be executed in different orders than are expressly illustrated or described, as appropriate for particular examples of the technology. Further, in some examples, certain operations can be executed in parallel or partially in parallel, including by dedicated parallel processing devices, or separate computing devices configured to interoperate as part of a large system.

(97) In some examples, aspects of the technology, including computerized implementations of methods according to the technology, can be implemented as a system, method, apparatus, or article of manufacture using standard programming or engineering techniques to produce software, firmware, hardware, or any combination thereof to control a processor, also referred to as an electronic processor, (e.g., a serial or parallel processor chip or specialized processor chip, a single- or multi-core chip, a microprocessor, a field programmable gate array, any variety of combinations of a control unit, arithmetic logic unit, and processor register, and so on), a computer (e.g., a

processor operatively coupled to a memory), or another electronically operated controller to implement aspects detailed herein. Accordingly, for example, examples of the technology can be implemented as a set of instructions, tangibly embodied on a non-transitory computer-readable media, such that a processor can implement the instructions based upon reading the instructions from the computer-readable media. Some examples of the technology may include (or utilize) a control device such as, e.g., an automation device, a special purpose or programmable computer including various computer hardware, software, firmware, and so on, consistent with the discussion herein. As specific examples, a control device may include a processor, a microcontroller, a field-programmable gate array, a programmable logic controller, logic gates etc., and other typical components that are known in the art for implementation of appropriate functionality (e.g., memory, the communication systems, power sources, user interfaces and other inputs, etc.).

(98) As used herein in the context of computer implementation, unless otherwise specified or limited, the terms “component,” “system,” “module,” “block,” and the like are intended to encompass part or all of computer-related systems that include hardware, software, a combination of hardware and software, or software in execution. For example, a component can be, but is not limited to being, a processor device, a process being executed (or executable) by a processor device, an object, an executable, a thread of execution, a computer program, or a computer. By way of illustration, both an application running on a computer and the computer can be a component. One or more components (or system, module, and so on) can reside within a process or thread of execution, can be localized on one computer, can be distributed between two or more computers or other processor devices, or can be included within another component (or system, module, and so on).

(99) In exemplary embodiments, the ingestion & rules engine **112**, the compromised identity set service **114**, the message bus **116**, and corresponding elements of FIGS. **1A** and **1B** may reside at a geographic location such as a country, state, county, city, or building or may reside in a cloud based architecture without a designated geographic location. In another exemplary embodiment, some of the ingestion & rules engine **112**, the compromised identity set service **114**, the message bus **116**, and other corresponding elements of FIGS. **1A** and **1B** may reside at one or more geographic locations while others reside in one or more cloud based architectures without departing from the spirit of the disclosure herein.

(100) Also as used herein, unless otherwise limited or defined, “or” indicates a non-exclusive list of components or operations that can be present in any variety of combinations, rather than an exclusive list of components that can be present only as alternatives to each other. For example, a list of “A, B, or C” indicates options of: A; B; C; A and B; A and C; B and C; and A, B, and C. Correspondingly, the term “or” as used herein is intended to indicate exclusive alternatives only when preceded by terms of exclusivity, such as, e.g., “either,” “only one of,” or “exactly one of.” Further, a list preceded by “one or more” (and variations thereon) and including “or” to separate listed elements indicates options of one or more of any or all of the listed elements. For example, the phrases “one or more of A, B, or C” and “at least one of A, B, or C” indicate options of: one or more A; one or more B; one or more C; one or more A and one or more B; one or more B and one or more C; one or more A and one or more C; and one or more of each of A, B, and C. Similarly, a list preceded by “a plurality of” (and variations thereon) and including “or” to separate listed elements indicates options of multiple instances of any or all of the listed elements. For example, the phrases “a plurality of A, B, or C” and “two or more of A, B, or C” indicate options of: A and B; B and C; A and C; and A, B, and C. In general, the term “or” as used herein only indicates exclusive alternatives (e.g., “one or the other but not both”) when preceded by terms of exclusivity, such as, e.g., “either,” “only one of,” or “exactly one of.”

(101) In the description above and the claims below, the term “connected” can refer to a physical connection or a logical connection. A physical connection indicates that at least two devices or systems co-operate, communicate, or interact with each other, and are in direct physical or

electrical contact with each other. For example, two devices are physically connected via an electrical cable. A logical connection indicates that at least two devices or systems co-operate, communicate, or interact with each other, but may or may not be in direct physical or electrical contact with each other. Throughout the description and claims, the term “coupled” may be used to show a logical connection that is not necessarily a physical connection. “Co-operation,” “the communication,” “interaction” and their variations include at least one of: (i) transmitting of information to a device or system; or (ii) receiving of information by a device or system.

(102) The word “exemplary” is used herein to mean serving as an example, instance, or illustration. Any aspect or design described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects or designs.

(103) Any mark, if referenced herein, may be common law or registered trademarks of third parties affiliated or unaffiliated with the applicant or the assignee. Use of these marks is by way of example and shall not be construed as descriptive or to limit the scope of disclosed or claimed embodiments to material associated only with such marks.

(104) Although the present technology has been described by referring to certain examples, workers skilled in the art will recognize that changes can be made in form and detail without departing from the scope of the discussion.

Claims

1. A system comprising: an ingestion & rules engine to: receive, electronically from a data breach set, breached data comprising compromised user information, receive, electronically from an identity system, account data comprising user account information, determine, by comparing the account data with the breached data, whether or not the user account information is found in the compromised user information, identify, when the ingestion & rules engine determines that the user account information is found in the compromised user information, the user account information as compromised, and output, in an information record when the ingestion & rules engine identifies the user account information as compromised, the user account information; a compromised identity set service to: store, in a database when the compromised identity set service receives the information record from the ingestion & rules engine, the user account information; and a message bus to: broadcast, electronically to a plurality of security subscription services when the compromised identity set service stores the user account information, a publication notifying the security subscription services that the user account information is compromised.
2. The system according to claim 1, wherein the user account information is compromised when appropriated without authorization by an owner of the user account information.
3. The system according to claim 1, wherein the ingestion & rules engine is to: calculate, when the ingestion & rules engine determines that the user account information is in the database, a risk score that quantifies an amount of risk.
4. The system according to claim 3, wherein the information record includes the risk score.
5. The system according to claim 1, wherein the compromised identity set service is to: receive, electronically from an external service, a query to ascertain whether or not the user account information is compromised.
6. The system according to claim 5, wherein the compromised identity set service is to: output, electronically to the external service when the compromised identity set service determines that the user account information is in the database, an acknowledgment that the user account information is compromised.
7. The system according to claim 5, wherein the external service is partner app shopping, a device service, or a security subscription service.
8. A method comprising: receiving, electronically by an ingestion & rules engine from a data breach set, breached data comprised of compromised user information; receiving, electronically by

the ingestion & rules engine from an identity system, account data comprised of user account information; determining, by the ingestion & rules engine in response to comparing the account data with the breached data, whether or not the user account information is found in the compromised user information; designating, by the ingestion & rules engine in response to determining that the user account information is found in the compromised user information, the user account information as compromised; and outputting, by the ingestion & rules engine in an information record in response to identifying the user account information as compromised, the user account information; storing, by a compromised identity set service in response to receiving the information record from the ingestion & rules engine, the user account information in a database; and broadcasting, electronically by a message bus to a plurality of security subscription services in response to the compromised identity set service storing the user account information, a publication notifying the security subscription services that the user account information is compromised.

9. The method according to claim 8, wherein the user account information is compromised when appropriated without authorization by an owner of the user account information.

10. The method according to claim 8, further comprising: calculating, by the ingestion & rules engine when the ingestion & rules engine determines that the user account information is in the database, a risk score that quantifies an amount of risk.

11. The method according to claim 10, wherein the information record includes the risk score.

12. The method according to claim 8, further comprising: receiving, by the compromised identity set service electronically from an external service, a query to ascertain whether or not the user account information is compromised.

13. The method according to claim 12, further comprising: outputting, electronically by the compromised identity set service to the external service when the compromised identity set service determines that the user account information is in the database, an acknowledgment that the user account information is compromised.

14. The method according to claim 12, wherein the external service is partner app shopping, a device service, or a security subscription service.

15. A non-transitory processor-readable memory storing instructions that, when executed by an electronic processor cause an ingestion & rules engine to: receive, electronically from an information distributor by wire or wirelessly via a communication network, a data breach set; extract, electronically from the data breach set, breached data comprised of compromised user information; receive, electronically from an identity system by wire or wirelessly via the communication network, account data comprised of user account information; and determine, in response to comparing the account data with the breached data, whether or not the user account information is found in the compromised user information.

16. The non-transitory processor-readable memory according to claim 15, wherein the instructions, when executed by the electronic processor cause the ingestion & rules engine to: output, to a compromised identity set service in response to determining that the compromised user information includes the user account information, the user account information in an information record.

17. The non-transitory processor-readable memory according to claim 15, wherein the instructions, when executed by the electronic processor cause the ingestion & rules engine to: designate, in response to determining an absence of the user account information in the compromised user information, the user account information as uncompromised.

18. The non-transitory processor-readable memory according to claim 15, wherein the breached data is information that is personal, sensitive, private, and/or proprietary.

19. The non-transitory processor-readable memory according to claim 15, wherein the compromised information may include usernames, passwords, addresses, credit card information, banking information, and/or any other information that is personal, sensitive, private, and/or proprietary.

20. The non-transitory processor-readable memory according to claim 15, wherein the ingestion & rules engine is to receive the data breach set from the information distributor intermittently.

21. The non-transitory processor-readable memory according to claim 15, wherein the ingestion & rules engine is to receive the data breach set from the information distributor on a schedule or a timetable.
